

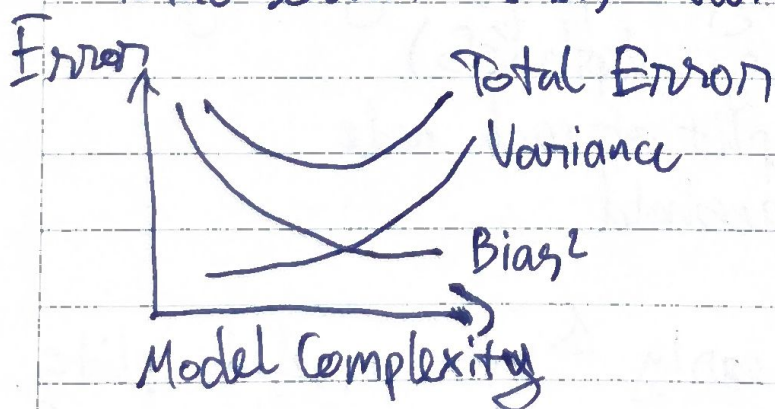
W01-04

Q1: Underfitting vs Overfitting

- Underfitting: when the model is too simple to capture the underlying pattern. Both ~~the~~ training error and test error are high. Ex: fitting a degree-1 polynomial to quadratic data.
- Overfitting: when the ~~model~~ model is too complex and memorizes noise in the training data. Train error is low but test error is high. Ex: fitting a degree-15 polynomial to 20 data point.
- Detection: A large gap between training errors & validation errors indicates overfitting. If both are high $\rightarrow$ underfitting

Q2: Bias-Variance ~~Trade~~ Tradeoff

- Bias: Error from oversimplifying assumptions. High bias $\rightarrow$ ~~overfitting~~ Underfitting.
- Variance: Error from sensitivity to specific training data. High variance $\rightarrow$ Overfitting.
- Total error: $\text{Bias}^2 + \text{Variance} + \text{Irreducible Noise}$



As model complexity increase:
 Bias $\downarrow$ (model more flexible)
 Variance $\uparrow$ (model more sensitive to noise)
 Optimal model minimizes total error: balancing bias & variance

Q3 Information gain

Set S with 14 samples: 9 Y, 5 N. Splitting attribute: ~~With~~ "Wind" (Weak / Strong)

$$\text{Entropy before split: } H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

Empire the love of learning

After split:

$$\text{Wind} = \text{Weak}: 8 \text{ samples } (6Y, 2N) \Rightarrow H = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} \\ = 0.811$$

$$\text{Wind} = \text{Strong}: 6 \text{ samples } (3Y, 3N) \Rightarrow H = 1.0$$

Information gain:

$$IG = H(S) - \frac{8}{14} H(\text{Weak}) - \frac{6}{14} H(\text{Strong})$$

$$= 0.940 - 0.811 \times \frac{8}{14} - \frac{6}{14} \times 1$$

$$= 0.048 > 0 \Rightarrow \text{split is useful.}$$

Compare IG across all attributes to choose the one with the highest gain as root / node split.

Q4 Why decision tree overfit & pruning

- A decision tree can grow until every leaf contain exactly 1 sample, effectively memorizing the entire training set including noise. Deeper trees have higher variance.

- Method 1: Pre-pruning. Stop growing using constraints:

- + Limit max-depth (e.g.: depth ≤ 5)

- + Require min-sample-split at each node

- + Stop if $IG < \text{threshold}$

- + Pro: fast & simple

- + Con: may stop too early & miss useful splits.

- Method 2: Post-pruning. Grow full tree & trim

- + For each internal node try replacing it with a leaf

- + If validation accuracy does not decrease $\rightarrow$ keep

- + Pro: generally produces better trees

- + Con: more computationally expensive.